

# METALLIC BONDING – DIAGRAM B

## Interpretation & Practice

Progress: ○ ○

Metallic bonding is the strong pull between **positive metal ions** (atoms that lost electrons) and **delocalised electrons** (electrons that move freely).

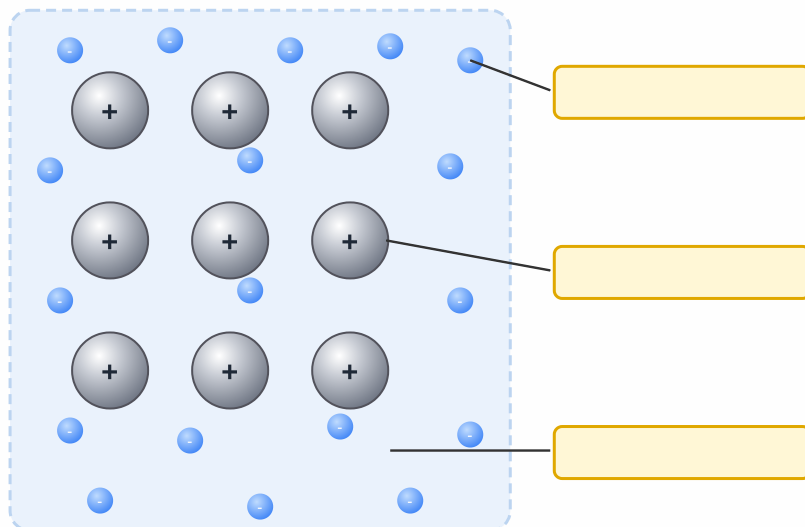
### 1 LABEL THE STRUCTURE

Look at the diagram below. Use the words in the box to fill in the labels.

delocalised electrons

positive metal ions

layers can slide



- ☐ Find the large circles with '+' signs — these are the metal ions.
- ☐ Find the small circles with '-' signs — these are the electrons.
- ☐ Think about what happens between the layers.

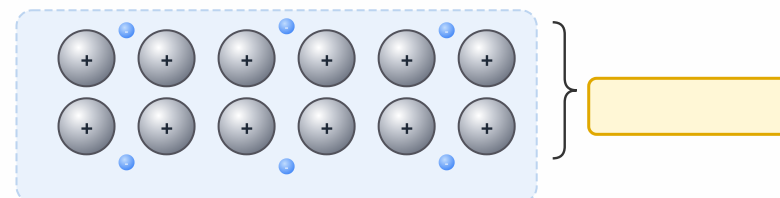
**TIP:** Positive metal ions are held in a 'sea' of delocalised electrons — like balls floating in water.

### 2 MALLEABILITY

The diagram shows layers of metal ions. When you push on a metal, the layers can slide over each other.

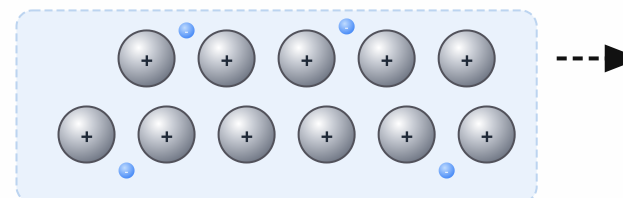
Explain why the layers of ions can slide over each other.

BEFORE FORCE



FORCE APPLIED

AFTER FORCE



- ☐ Look at the 'before' and 'after' diagrams. What changed?
- ☐ Think: are the ions bonded to specific other ions, or held by a general 'sea' of electrons?
- ☐ Because the electrons are everywhere (delocalised), the ions can move without bonds breaking.
- ☐ Write your answer using the words: delocalised, electrons, slide, ions.

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Think: What property of metals does this structure explain? (Hint: can you bend a metal?)